CAP3032: Final Project - I
Game Design Document: Rhythm Mania
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Game Design

We are designing a rhythm game where circles will appear on the screen with directional arrows inside them. The objective is for the user to click the correct arrow key that corresponds to the circles.

Game Complexity

The game will have two difficulties:

- Easy: Slower song
- Difficult: Faster song

User's Manual (User's manual explaining how to play the game)

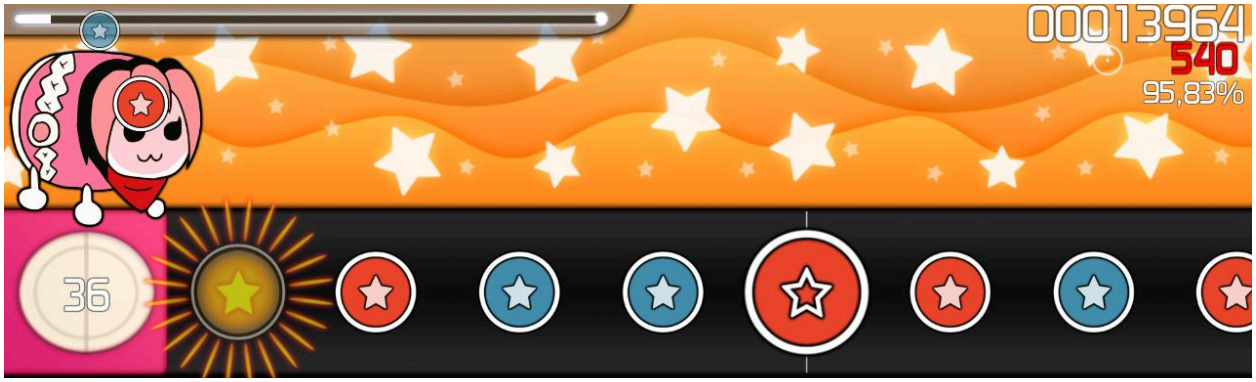
The user's manual explains the game to players. This will be a button that users can access in the main menu before they proceed to start the game (after clicking the start button). Here, they will be given the following instructions to better understand the logistics of the game and the point system.
How to Play:

In the menu screen, press E to play in Easy mode, and press H to play in Hard mode. Press M to open the user's manual, and press Q to quit the game.
Easy: 100 beats per minute song, less frequent beats.
Hard: 160 beats per minute song, more frequent beats.

Circles with keys will appear onscreen, with each corresponding to a beat for the song chosen. Press the correct arrow key as soon as the key circle is on top of the yellow-outlined reference circle.
You earn points based on how close your keystrokes were from completely overlapping the reference circle; you can either score a 'Perfect,' a 'Good,' a 'Poor,' or a 'Miss.'
The points accumulate and assign you a grade based off how well you performed throughout the song, as follows: S (excellent), A, B, C, F (poor)

User Interface :

This is an inspiration for the user interface that we plan to implement.



Menu

- How to Play Button
  - Which will link to the user's manual that explains how to play the game
- Start Button
  - Start the game
- Select Difficulty Button
  - Easy Mode
  - Hard mode
- Quit Button
  - Exit the Game at any time
- Credits/References Button
  - Credits to the amazing authors and references used to create the game

Interaction

- Mouse Control: Users can click the buttons in the menu (e.g. to start, quit, manual)
- Keyboard Control: Users will use the arrow keys on their keyboards to match the incoming circles on the screen that go with the song

Sound (Sound: background music, and at least one other sound which plays when certain event, such as hitting a ball, happens)

- Song 1: --- SHINee Replay
  - https://mp3juice.sc/
- Song 2: --- PinkPantheress True Romance
  - https://mp3juice.sc/
- Sound Effect 1- Perfect: --- (each time the user gets a perfect hit)
  - https://freesound.org/s/538147/

    - Cut version: https://limewire.com/d/GW8xC#VgTn0n5rzf
- Sound Effect 2 - User Clicks: --- (each time the user clicks the arrows)
    - https://freesound.org/s/789793/
- Menu Background Song - Perfume Perfume
    - https://mp3juice.sc/

Images

- Miku art
- Background photo during the game, low opacity
- Small images to give the game more personality

Code Structure

Function	Description
void setup()	Initial setup for all our static variables
void draw()	Main display for screen appearance (calls other functions like Menu and game)
menu()	Text + button display for the different screens you can navigate to
Game class	Class for holding all functionalities within the game. This includes game, checkInput, and spawnNotes functions.
userManual()	Text display + image examples for how to play the game
game()	Main Game play
checkInput()	arrow detection
spawnNotes()	Timing system for notes Circles/arrows falling or appearing Detect arrow key presses Check timing accuracy (perfect/good/miss)

	Rather than syncing the music which may be more challenging, a simpler approach is: - Predefining when notes appear (by using an array of timestamps) - Compare millis() to note timing - Allow a small “hit window” (e.g., ± 200ms)
Note class	Helper class to manage individual notes scrolling past on screen. Functions: constructor (int dir, float trackY) void update() - updates each note so that they rapidly move left. void display() - displays note as circle onscreen. Red if it’s a left/right arrow key and Blue if it’s an up/down arrow key.